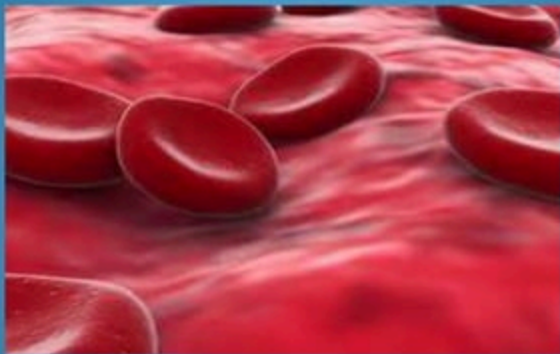


Blood Transfusions



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M.Sc. Nursing 1st year



Overview

- What is a blood transfusion?
- Purpose
- Indications
- Blood products
- Nursing implications
- Administering a blood transfusion (skill)



Introduction

Transfusion of blood and its products is very common medical procedure in the practice of modern medicine. Nursing staff plays a very important role and crucial role in the blood transfusion procedure as they are called upon to procure and blood products from the blood transfusion laboratory and conduct supervise the transfusion.



Definition

- Blood transfusion is the transfusion of the whole blood or its component such as blood cells or plasma from one person to another person.
- Reaches patient's blood vessels and enters the circulatory system

Purpose

- Restore blood volume
- Replace clotting factors
- Improve oxygen carrying capacity
- Restore blood elements that are depleted
- Prevent complications
- To raise the haemoglobin level
- To provide antibodies

BLOOD PRODUCTS

BLOOD- Blood is the life-maintaining fluid that circulates through the body.

- Components of the blood which are collected from a donor for use in blood transfusion

PRODUCTS

- Most common type of blood product for transfusion
- Used to increase the oxygen-carrying capacity of blood
- Help the body get rid of carbon dioxide and other waste products
- 1 unit of PRBCs = raises hematocrit by 2-3%

Fresh Frozen Plasma (FFP)

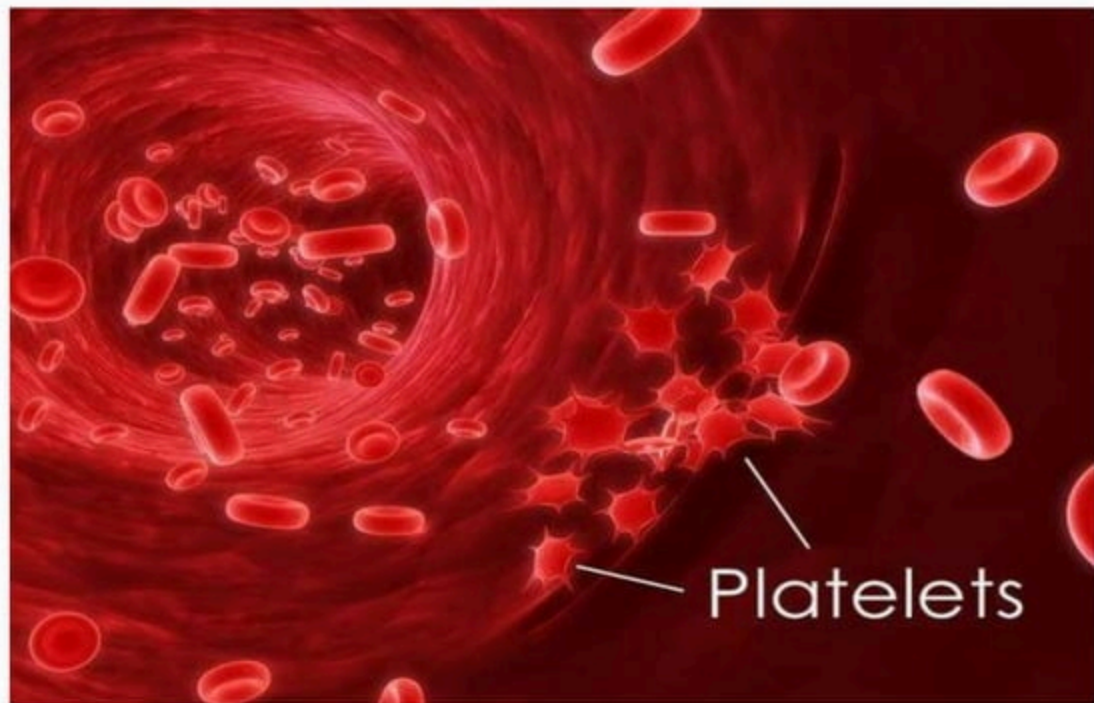
- Plasma is the liquid component of blood; it has proteins called *clotting factors*
- Expands blood volume and provides clotting factors
- Contains no RBCs
- 1 unit of FFP = increases level of any clotting factor by 2-3%

Fresh Frozen Plasma



Platelets

- Also known as *thrombocytes*
- Tiny cell structures necessary in blood clotting process
- Replaces platelets in clients with bleeding disorders, or platelet deficiency
- 1 unit = increases the average adult client's platelet count by about **5,000 platelets/microliter**



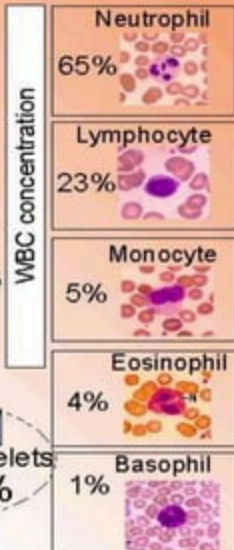
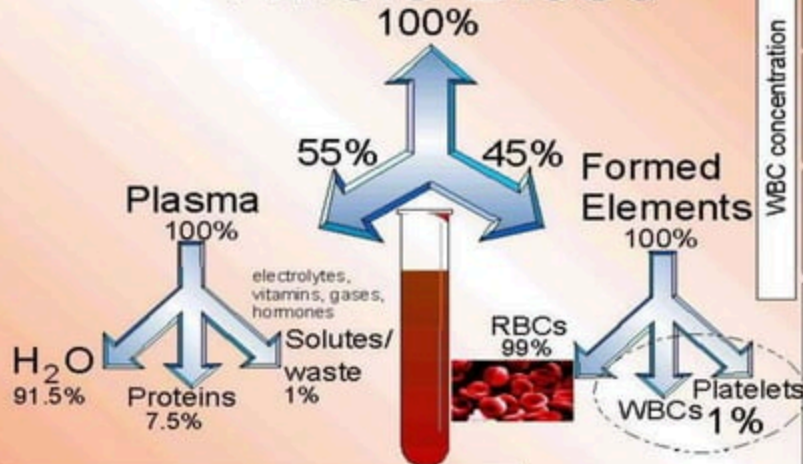
Platelets



Whole Blood

- Not commonly used except for extreme cases of acute hemorrhage
- Replaces blood volume and all blood products

Whole Blood



Plasma Protein composition

Albumins 54% smallest, most abundant, transport fatty acids	Globulins 38% from plasma & liver cells, 38% Gamma Globulins Immunoglobulins	Fibrinogen 7% + trace substances
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fibrinogen- 7%, made by liver

Autologous Red Blood Cells

- Used for blood replacement following planned elective surgery
- Must be donated 4-5 weeks prior to surgery

Albumin and Plasma Protein Fraction

- Blood volume expander
- Provides plasma protein

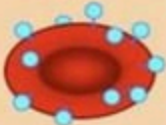
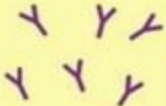
Clotting Factors and Cryoprecipitate

- A portion of plasma containing certain specific clotting factors
- Used for clients with clotting factor deficiencies
- Contains Fibrinogen

FUNCTIONS OF BLOOD

- Blood carries the following to the body tissues:
 - Nourishment.
 - Electrolytes.
 - Hormones.
 - Vitamins.
 - Antibodies.
 - Heat
 - Oxygen.
- Blood carries the following away from the body tissues:
 - Waste matter.
 - Carbon dioxide.

Blood Grouping

ABO Blood Groups				
Antigen (on RBC)	Antigen A 	Antigen B 	Antigens A + B 	Neither A or B 
Antibody (in plasma)	Anti-B Antibody 	Anti-A Antibody 	Neither Antibody	Both Antibodies 
Blood Type	Type A Cannot have B or AB blood Can have A or O blood	Type B Cannot have A or AB blood Can have B or O blood	Type AB Can have any type of blood Is the universal recipient	Type O Can only have O blood Is the universal donor

The Rhesus system

- The red blood cell membrane antigen important here is the Rhesus (Rh) antigen, or Rhesus factor. About 85% of people have this antigen; they are Rhesus positive (Rh+) and not therefore make anti-Rhesus antibodies. The remaining 15% have no Rhesus antigen (they are Rhesus negative, or Rh-) Rh- individuals are capable of making anti Rhesus antibodies, but are stimulated to do so only in certain circumstances, e.g. in pregnancy, or as the result of an incompatible blood transfusion.

GENERAL INSTRUCTIONS FOR GIVING BLOOD TRANSFUSIONS

- Donor shall be free of diseases
- No history of any diseases
- They have not donated blood within the previous 90 days. (3 months).
- They should be physically healthy, and should be between 18 and 65 years.
- Donor must have a normal temperature, pulse and blood pressure.
- They must not have been pregnant within the last 6 months.
- Their haemoglobin level must be above 12 grams per 100ml.

Continue

- The donors are disqualified who have a history of recent dental surgery or major surgery, receipt of blood or blood components, immunizations or vaccinations, use of narcotics etc.
- Before the blood transfusion, the ABO grouping and Rh typing with the recipient's blood should be done
- Before the blood is transfused, the donor's blood must be cross matched with the recipient's serum (plasma) and the recipient's red blood cells

Continue

- Explain the procedure to the donor and reassure him/her to with confidence and co-operation.
- Blood should not be collected to the empty stomach.
- Before the donor is allowed to leave the hospital, his pulse rate should be questioned as to have any giddiness.
- A second withdrawal of blood should not be made until the blood volume and constitutes have returned to normal, that is usually after 3 months.

COLLECTION, STORAGE AND TRANSPORTATION OF BLOOD

- Collection of blood from the donor is done in the laboratory by the laboratory techniques. The donor's blood collected into a sterile container containing anticoagulant solution. The anticoagulant used is ACD solution (acid citrate dextrose) or citrate phosphate dextrose.
- All the articles used for the collection of blood should sterile. They should be pyrogen free.

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- Each donor unit must be labeled in clear, readable letters, bearing the following information to be verified at the time of administration:
 - Name of the donor.
 - Donor number.
 - ABO grouping
 - Rh typing
 - Date of drawing blood
 - Date of expiry
 - Results of tests for hepatitis and syphilis

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- The donor blood, immediately after it is withdrawn, should be placed in the refrigerator. Usually it is stored temperature of 1 to 6 degree centigrade.
 - Stored blood shall be inspected daily and before use evidence of hemolysis or bacterial contamination.
 - The transportation of the blood in the hospital should be done within 30 minutes after it is taken from the place of storage.
 - Freezing and heating of the blood will destroy the blood cells.

REGARDING ADMINISTRATION OF BLOOD TO THE RECIPIENT

- When sending the recipient blood sample for grouping and cross matching, it must be carefully labeled at the bedside of the recipient
- A request form should accompany the specimen and it should contain the following data :
 - The recipient's name
 - Hospital number
 - Bed number
 - Ward number
 - Name of the physician
 - Exact amount of blood component requested.
 - Diagnosis of the patient
 - Any blood transfusion given earlier,

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- It is essential that the physician writes all orders for typing, cross matching and administration of whole blood and blood products.
 - When blood or blood products have to be administered, it is recommended that two registered nurse or a physician and a registered nurse should independently verify all identifying information on the report of the cross match, unit label and patient's identifications. Any error in the identification, results in the major adverse reactions. Should there be any discrepancy, the unit should be returned to the blood bank with the remarks.

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- Whole blood and the blood products should be administered through an appropriate, sterile, pyrogen free transfusion set containing a filter which will remove clots and larger aggregates of leucocytes and platelets.
 - Care is to be taken to prevent introduction of air into the apparatus.
 - It is recommended to use 18 gauge needle for infusion, to prevent damage to the red cells and provide an adequate rate of flow.

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- No medication- antibiotics, vitamins, calcium, etc. should be added to the unit of blood or administered through the same intravenous system as they may cause damage to the red cells.
 - If I.V infusion is to be given immediately before, during or after the blood transfusion, always use physiologic saline to prevent haemolysis of the blood in the tubing. Dextrose should be avoided. If another I.V. fluid is to be given, rinse the infusion set with normal saline before starting the solution

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- Prior to administer of blood, the patient's vital signs should be recorded correctly on the nurse's record to provide a baseline for further observation.
 - Adjust the rate of flow to 5 to 10 ml per minute during the first 30 minutes of transfusion, to detect any complication as early as possible. Give the blood at a slower rate if the patient is elderly, suffering from heart disease and lung disease, anemia, debilitating diseases etc.

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- Whole blood and packed cells administered cold. No attempt is made to heat the blood. However, blood may be allowed to stand at the room temperature for 30 to 45 minutes before it is administered to the patient.
 - Once the blood is exposed to the atmosphere (the unit is opened), it should be discarded.

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- The procedure involved in administration of the blood is the same as that of administration of I.V. infusions. Watch the patient carefully for the onset of any complications. Any reaction developed in the patient should be reported to the charge nurse and physician immediately.



The following observation are made throughout the procedure :

- Rate of flow
- Signs of circulatory overloaded
- Urinary output (
- The needle site of signs of infiltration, hematoma, and dislodgement of needle etc.
- Blood level in the container. Never allow the blood bottle to be completely empty to prevent the entry of air.
- Reaction to the blood transfusion
- Patency of the infusion set

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- Keep the patient warm and comfortable with blankets if necessary.
 - Offer bedpan before starting the procedure and as necessary.



Record on the nurse's record with date and time:

- The amount of blood administered.
- The group and the type of blood administered.
- The rate of flow.
- Any reaction observed.
- Any medication administered e.g., antihistamines.



When leaving the ward, the charge nurse should report the following to the relieving nurse.

- The name and the bed number of the patient receiving the blood transfusions.
- The time at which the drip has started
- The type of blood that is given
- Amount of blood that is administered and the quantity to be given more
- Any specific precautions to be followed
- The specific rate of flow
- Any reactions observed
- Any medications administered
- General condition of the patient

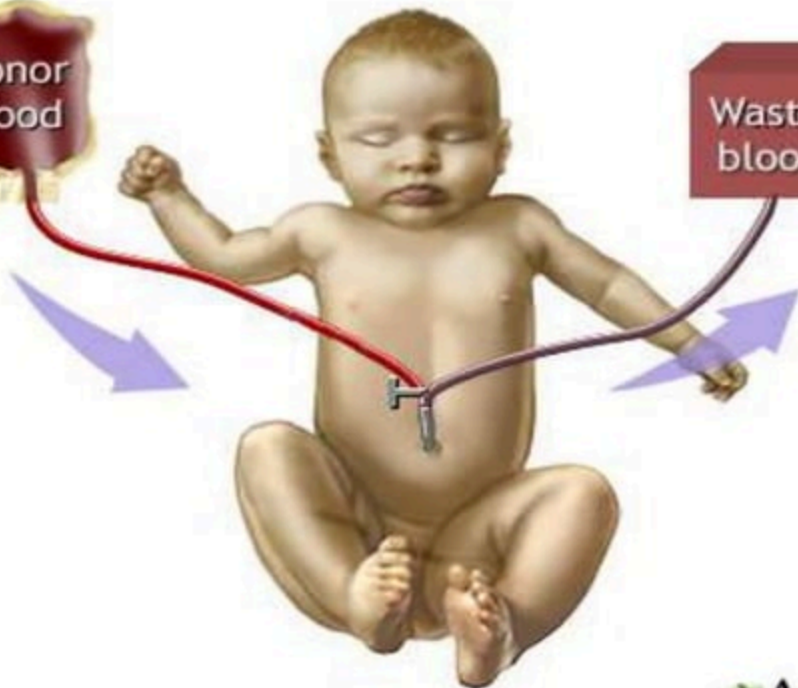


COMPLICATIONS/ REACTIONS OF BLOOD TRANSFUSION



EXCHANGE BLOOD TRANSFUSION

An exchange transfusion is a medical procedure in which your blood is removed and replaced with plasma or donor blood. This is done via a catheter. The procedure is used to save the life of an adult or child with life-threatening blood abnormalities.





Why Are Exchange Transfusions Done?

An exchange transfusion reverses or counteracts the symptoms of jaundice or other blood diseases, such as sickle cell anemia.

Risks of an Exchange Transfusion

These risks include:

- Mild allergic reactions
- Fever due to infection
- Trouble breathing
- Anxiety
- Electrolyte abnormalities
- Nausea
- Chest Pains

PREPARATION FOR THE TEST

- Before your transfusion, give a simple blood test to confirm blood type and then prick finger with a small needle to get a few drops of blood.
- Blood will then be labelled and sent to a lab where a machine will analyze it to determine blood type. This ensures that the blood receive by transfusion is a match for blood type. If the donor blood is not a match, it will make sick to the patient.
- To check allergic reactions to blood transfusions in the past.

GENERAL INSTRUCTIONS

- Explain the procedure to the relative and reassure them.
- The blood should cross match properly.
- The blood should be as fresh as possible and it should never be more than 5 days.
- Once the transfusion will complete, check blood pressure, heart rate, and temperature.
- The venous pressure should be assessed from time to time and careful watch is kept against overloading.
- All apparatus should be flushed from time to time with heparinised saline.

PROCEDURE OF BLOOD TRANSFUSION

Articles Required

1. A tray containing
 - Sprit
 - Iodine
 - Tr . benzoin
 - Cotton swab in container
 - Splint and bandage

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- Tourniquet
 - Mackintosh with treatment towel
 - Sterile blood transfusion set with filter
 - Large needle
 - Kidney tray and paper bag
 - Elastoplasts or adhesive plaster and scissors
 - Gloves
 - Bottle cover
 - Normal saline bottle

2. I.V. stand

PROCEDURE

- Explain the procedure to patient. Restraint children, select a site (basalic and cephalic, veins, saphenous)
- Wash hands. Remove the bottle seal from top, clean the top with spirit swab, holding the bottle upright, insert the drip set and air vent into the bottle
- Close the clamp and hang the bottle on the I.V. stand about 18-25 high
- Close the clamp and hang the bottle on the I.V. tubing. Open the clamp and flush the I.V. fluid through the tubing and needle into the kidney tray until air is removed. Clamp the tubing again, apply protective cap over the needle

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- Prepare few strips of adhesive tapes
 - Site preparation: Apply a tourniquet firmly 6 to 8" proximal to the site
 - Encourage the patient to clench and unclench the fist rapidly. Lightly tap the vein with finger tips
 - Clean the area with a spirit swab
 - Insert the needle into the vein at 15-30 degree angle and once it enters the vein, make it parallel with the skin and follow the course of vein

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- When back flow of blood occurs into the needle and tubing, insert the needle further up into the vein about $\frac{3}{4}$ to 1"
 - Release the tourniquet and open the clamp to let the fluid run
 - Secure the needle and tubing by adhesive strips
 - Immobilize with splint
 - Record in nurse's notes and intake output chart
 - Change the bottle whenever it finishes with prescribed I.V. fluid

THANK



YOU

